

13 Functions of Several Variables

Chain Rule (Two Independent Variables). Let z be a differentiable function of x and y , where x and y are differentiable functions of s and t . Then

$$\frac{dz}{dt} = \frac{dz}{dx} \frac{dx}{dt} + \frac{dz}{dy} \frac{dy}{dt}.$$

Gradient (Two Dimensions). Let f be differentiable at the point (x, y) . The **gradient** of f at (x, y) is the vector-valued function

$$\nabla f(x, y) = \langle f_x(x, y), f_y(x, y) \rangle = f_x(x, y)\mathbf{i} + f_y(x, y)\mathbf{j}.$$

Directional Derivative. Let f be differentiable at (a, b) and let $\mathbf{u} = \langle u_1, u_2 \rangle$ be a unit vector in the xy -plane, the **directional derivative of f at (a, b) in the direction of $\mathbf{u}$** is

$$D_{\mathbf{u}}f(a, b) = \langle f_x(a, b), f_y(a, b) \rangle \cdot \langle u_1, u_2 \rangle$$

$$D_{\mathbf{u}}f(a, b) = \nabla f(a, b) \cdot \mathbf{u}.$$

Second Derivative Test. Suppose that the second partial derivatives of f are continuous throughout an open disk centered at the point (a, b) , where $f_x(a, b) = f_y(a, b) = 0$. Let $D(x, y) = f_{xx}f_{yy} - f_{xy}^2$.

1. If $D(a, b) > 0$ and $f_{xx}(a, b) < 0$, then f has a local maximum value at (a, b) .
2. If $D(a, b) > 0$ and $f_{xx}(a, b) > 0$, then f has a local minimum value at (a, b) .
3. If $D(a, b) < 0$, then f has a saddle point at (a, b) .
4. If $D(a, b) = 0$, then the test is inconclusive.

Helpful formulas

1. $\frac{d}{dx}(x^n) = nx^{n-1}$
2. $\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$ (Chain rule)
3. $\frac{d}{dx}(f(x)g(x)) = f'(x)g(x) + f(x)g'(x)$
4. $\frac{d}{dx}e^x = e^x$
5. $\frac{d}{dx}\sin ax = a \cos ax$
6. $\frac{d}{dx}\cos ax = -a \sin ax$
7. $\int x^n dx = \frac{1}{n+1}x^{n+1} + C; n \neq -1$
8. $\int \cos ax dx = \frac{1}{a}\sin ax + C$
9. $\int \sin ax dx = -\frac{1}{a}\cos ax + C$
10. $\int \cos^2 x dx = \frac{x}{2} + \frac{\sin 2x}{4} + C$
11. $\int \sin^2 x dx = \frac{x}{2} - \frac{\sin 2x}{4} + C$

